

CentreCOM® GS980EM Series

Gigabit Layer 3 Lite PoE++/PoE Pass-through Switches

The Allied Telesis GS980EM Series of Gigabit Layer 3 Lite PoE++/PoE pass-through switches offer an impressive set of features in a compact design. Ideal for deployment at the network edge, the GS980EM Series feature flexible Power over Ethernet capabilities to support IoT device connectivity in today's converged business environments.

Overview

Allied Telesis GS980EM Series are secure and reliable, offering 8 x Gigabit PoE enabled ports and 2 x SFP uplinks, and providing a high value solution for flexible PoE at the network edge.

The GS980EM/10H can provide up to 90 Watts (PoE++) on all ports. This enables powering high power devices such as high resolution PTZ cameras with heater/blowers for outdoor applications, enhanced infrared lighting and lighting controllers, and more. The GS980EM/10H requires a PWR300 external power supply to operate and provide PoE power. Up to three PWR300 PSUs can be used to increase the available PoE power.

The GS980EM/11PT can supply up to 30 Watts (PoE+) to connected devices. It can be powered by an AC power adapter, or by PoE¹. When deployed together, the GS980EM/11PT can be powered by the GS980EM/10H, while PoE pass-through enables power from the GS980EM/10H to pass through the GS980EM/11PT to power connected end points.

SPECIFICATIONS

Performance

- 10KB L2 jumbo frames
- 4094 configurable VLANs
- Up to 16K MAC addresses
- Up to 2K multicast entries
- 512 MB DDR3 SDRAM
- 128 MB NAND flash memory
- Packet buffer memory: 1.5MB

Reliability

- Modular AlliedWare Plus operating system
- Full environmental monitoring of PSUs, fans, temperature and internal voltages.
- MTBF (Mean Time Between failures)
 - GS980EM/10H: 400,000 hours
 - GS980EM/11PT: 240,000 hours

Flexibility and Compatibility

- 1G-SFP ports on GS980EM will support any combination of Allied Telesis 100Mbps and 1000Mbps SFP modules listed in this document under Ordering Information
- Port speed and duplex configuration can be set manually or by auto-negotiation

Diagnostic Tools

- Active Fiber Monitoring detects tampering on optical links
- Built-In Self test (BIST)
- Cable fault locator (TDR)
- Find-me device locator
- Automatic link flap detection and port shutdown
- Optical Digital Diagnostic Monitoring (DDM)
- Ping polling for IPv4 and IPv6

- Port mirroring
- Trace Route for IPv4 and IPv6
- Uni-Directional Link Detection (UDLD)

IPv6 Features

- RIP, OSPF, and Static routing for IPv4
- Device management over IPv6 networks with SNMP, Telnet, SSH
- IPv6 hardware ACLs
- Log to IPv6 hosts with Syslog
- IPv6 Ready certified
- DHCPv4 and v6 client
- Equal Cost Multi Path (ECMP) routing

Management

- Autonomous Management Framework Plus (AMF Plus) enables powerful centralized management and zero-touch device installation and recovery
- Manage the GS980EM with Vista Manager EX, our graphical single-pane-of-glass monitoring and management tool for AMF Plus networks. Also supports wireless and third party devices
- From AW+ 5.5.2-2, an AMF Plus license operating in the network provides all standard AMF network management and automation features, and also enables the AMF Plus intent-based networking features in Vista Manager EX (from version 3.10.1 onwards)
- AMF Security (AMF-Sec) enables a self-defending network—managing the GS980MX Series (or other AMF Plus switches) to automatically block the spread of malware by quarantining suspect end user devices
- IPFIX exports IP flow data in a network for analysis, so administrators have information for accounting, capacity planning, and optimization



Key Features

- Autonomous Management Framework™ Plus (AMF Plus) edge node
- Vista Manager EX compatible
- AMF-Security compatible
- Full 30 Watts of PoE+
- Up to 90 Watts of PoE++ (GS980EM/10H only)
- PoE pass-through (GS980EM/11PT only)
- AlliedWare Plus Enterprise-class operating system
- Energy Efficient Ethernet saves power
- Fanless design for silent operation
- Active Fiber Monitoring
- EPSRing™ enables resilient high-speed rings
- Static routing, RIP, OSPFv2
- IEEE 802.1x/MAC/Web authentication support
- IEEE 802.3x Flow Control
- Flexible deployment options including DIN rail mounting
- NETCONF/RESTCONF with YANG data modelling
- IPFIX (IP Flow Information Export)
- Enhanced Transmission Selection (ETS)

- Console management port on the front panel for ease of access
- Eco-friendly mode allows ports and LEDs to be disabled to save power
- Web-based Graphical User Interface (GUI)
- Industry-standard CLI

¹ The GS980EM/11PT uses PD port 11 to receive PoE power, but cannot be powered by PoE if the AC adapter is connected

Product Specifications

	10/100/1000T (RJ-45) PoE Enabled Ports	10/100/1000T (RJ-45) PoE-in-Port	1000X SFP Ports	Switching Fabric	Forwarding Rate
GS980EM/10H	8	-	2	24Gbps	14.96Mpps
GS980EM/11PT	8	1	2	24Gbps	16.4Mpps

Physical Specifications

	Width	Depth	Height	Mounting	Unpacked Weight	Packaged Weight	Packaged Dimensions
GS980EM/10H	210 mm (8.26 in)	180 mm (7.08 in)	42.5 mm (1.67 in)	Rack-mount	1.60 kg (3.53 lb)	2.7 kg (5.95 lb)	417 x 336 x 151 mm (16.42 x 13.23 x 1.67 in)
GS980EM/11PT	210 mm (8.26 in)	180 mm (7.08 in)	42.5 mm (1.67 in)	Rack-mount	1.60 kg (3.53 lb)	3.5 kg (7.71 lb)	417 x 336 x 151 mm (16.42 x 13.23 x 1.67 in)

Power and Noise Characteristics

	Maximum PoE Power	Maximum PoE Ports Supported					No PoE Load		Full PoE Load	
		PoE (7.5W)	PoE (15.4W)	PoE+ (30W)	PoE++ (60W)	PoE++ (90W)	Max Power Consumption	Max Heat Dissipation	Max Power Consumption	Max Heat Dissipation
GS980EM/10H	240W (1 x PWR300 PSU)	8	8	8	4	2	21W	71W	320W	218 BTU/h
	480W (2 x PWR300 PSUs)	8	8	8	8	5			600W	409 BTU/h
	720W (3 x PWR300 PSUs)	8	8	8	8	8			880W	600 BTU/h
GS980EM/11PT	0W (switch powered by 30W PoE) ¹	0	0	0	0	0	22W	75W	98W (using AC power adapter)	350W (using AC power adapter)
	31.6W (switch powered by 60W PoE) ¹	4	2	1	0	0				
	46.2W (switch powered by 90W PoE) ¹	6	3	1	0	0				
	62W (switch powered by AC Adaptor)	8	4	2	0	0				

¹ The GS980EM/11PT uses PD port 11 to receive PoE power, but cannot be powered by PoE if the AC adapter is connected

Latency (microseconds)

	Port Speed	
	100Mbps	1Gbps
GS980EM/10H	5.4µs	3.0µs
GS980EM/11PT	5.5µs	3.0µs

- NETCONF/RESTCONF northbound interface with YANG data modelling
- Powerful CLI scripting engine
- Comprehensive SNMP MIB support for standards based device management
- Built-in text editor
- USB interface allows software release files, configurations and other files to be stored for backup and distribution to other devices
- Event-based triggers allow user-defined scripts to be executed upon selected system events
- Wirespeed forwarding

Quality of Service

- 8 priority queues with a hierarchy of high priority queues for real time traffic, and mixed scheduling, for each switch port
- Limit bandwidth per port or per traffic class down to 64kbps
- Wire speed traffic classification with low latency essential for VoIP and real-time streaming media applications
- IPv6 QoS support

- Policy-based QoS based on VLAN, port, MAC and general packet classifiers
- Policy-based storm protection
- Extensive remarking capabilities
- Queue scheduling options for Strict priority, weighted round robin or mixed scheduling
- Type of Services (ToS) IP precedence and DiffServ marking based on layer 2, 3 and 4 headers
- Enhanced Transmission Selection (ETS)

Resiliency Features

- Control Plane Prioritization (CPP) ensures the CPU always has sufficient bandwidth to process network control traffic
- Dynamic link failover (host attach)
- EPSRing (Ethernet Protection Switched Rings) with Super-Loop Protection (SLP) and enhanced recovery for extra resiliency
- Loop protection: loop detection and thrash limiting
- PVST+ compatibility mode
- STP root guard

Security Features

- Access Control Lists (ACLs) based on layer 3 and 4 headers
- Dynamic ACLs assigned via port authentication
- ACL Groups enable multiple hosts/ports to be included in a single ACL, reducing configuration
- Configurable ACLs for management traffic
- Configurable auth-fail and guest VLANs

- Authentication, Authorization and Accounting (AAA)
- Bootloader can be password protected for device security
- BPDU protection
- DHCP snooping, IP source guard and Dynamic ARP Inspection (DAI)
- Dynamic VLAN assignment
- DoS attack blocking and virus throttling
- MAC address filtering and MAC address lock-down
- Network Access and Control (NAC) features manage endpoint security
- Port-based learn limits (intrusion detection)
- Private VLANs provide security and port isolation for multiple customers using the same VLAN
- Secure Copy (SCP)
- Secure File Transfer (SFTP) client
- Strong password security and encryption
- Simple Certificate Enrollment Protocol (SCEP) supports secure management
- Tri-authentication: MAC-based, web-based and IEEE 802.1x
- Web-based authentication

Environmental Specifications

- Operating temperature range: 0°C to 50°C (32°F to 122°F)
- Storage temperature range: -25°C to 70°C (-13°F to 158°F)
- Operating relative humidity range: 5% to 90% non-condensing

- Storage relative humidity range:
5% to 95% non-condensing
- Operating altitude:
3,000 meters maximum (9,843 ft)

Electrical Approvals and Compliances

- EMC: EN55032 class A, FCC class A, VCCI class A, ICES-003 class A
- Immunity: EN55024, EN61000-3-levels 2 (Harmonics), and 3 (Flicker) – AC models only

Safety

- Standards: UL60950-1, CAN/CSA-C22.2 No. 60950-1-03, EN60950-1, AS/NZS 60950.1
- Certification: UL, cUL

Restrictions on Hazardous Substances (RoHS) Compliance

- EU RoHS compliant
- China RoHS compliant

STANDARDS & PROTOCOLS

Cryptographic Algorithms

FIPS Approved Algorithms

Encryption (Block Ciphers):

AES (ECB, CBC, CFB and OFB Modes)

3DES (ECB, CBC, CFB and OFB Modes)

Block Cipher Modes:

CCM

CMAC

GCM

XTS

Digital Signatures & Asymmetric Key Generation:

DSA

ECDSA

RSA

Secure Hashing:

SHA-1

SHA-2 (SHA-224, SHA-256, SHA-384, SHA-512)

Message Authentication:

HMAC (SHA-1, SHA-2(224, 256, 384, 512)

Random Number Generation:

DRBG (Hash, HMAC and Counter)

Non FIPS Approved Algorithms

RNG (AES128/192/256)

DES

MD5

Ethernet Standards

IEEE 802.2 Logical Link Control (LLC)

IEEE 802.3 Ethernet

IEEE 802.3ab1000BASE-T

IEEE 802.3af Power over Ethernet (PoE)

IEEE 802.3at Power over Ethernet up to 30W (PoE+)

IEEE 802.3bt Power over Ethernet up to 90W (PoE++)

IEEE 802.3azEnergy Efficient Ethernet (EEE)

IEEE 802.3u 100BASE-X

IEEE 802.3x Flow control - full-duplex operation

IEEE 802.3z 1000BASE-X

IPv4 Features

- RFC 768 User Datagram Protocol (UDP)
- RFC 791 Internet Protocol (IP)
- RFC 792 Internet Control Message Protocol (ICMP)
- RFC 793 Transmission Control Protocol (TCP)
- RFC 826 Address Resolution Protocol (ARP)
- RFC 894 Standard for the transmission of IP datagrams over Ethernet networks
- RFC 919 Broadcasting Internet datagrams
- RFC 922 Broadcasting Internet datagrams in the presence of subnets
- RFC 932 Subnetwork addressing scheme
- RFC 950 Internet standard subnetting procedure

- RFC 951 Bootstrap Protocol (BootP)
- RFC 1027 Proxy ARP
- RFC 1035 DNS client
- RFC 1042 Standard for the transmission of IP datagrams over IEEE 802 networks
- RFC 1071 Computing the Internet checksum
- RFC 1122 Internet host requirements
- RFC 1191 Path MTU discovery
- RFC 1256 ICMP router discovery messages
- RFC 1518 An architecture for IP address allocation with CIDR
- RFC 1519 Classless Inter-Domain Routing (CIDR)
- RFC 1542 Clarifications and extensions for BootP
- RFC 1591 Domain Name System (DNS)
- RFC 1812 Requirements for IPv4 routers
- RFC 1918 IP addressing
- RFC 2581 TCP congestion control

IPv6 Features

- RFC 1981 Path MTU discovery for IPv6
- RFC 2460 IPv6 specification
- RFC 2464 Transmission of IPv6 packets over Ethernet networks
- RFC 3484 Default address selection for IPv6
- RFC 4007 IPv6 scoped address architecture
- RFC 4193 Unique local IPv6 unicast addresses
- RFC 4291 IPv6 addressing architecture
- RFC 4443 Internet Control Message Protocol (ICMPv6)
- RFC 5014 IPv6 socket API for source address selection
- RFC 5095 Deprecation of type 0 routing headers in IPv6

Management

AT Enterprise MIB including AMF Plus MIB and SNMP traps

Optical DDM MIB

SNMPv1, v2c and v3

IEEE 802.1AB Link Layer Discovery Protocol (LLDP)

RFC 1155 Structure and identification of management information for TCP/IP-based Internets

RFC 1157 Simple Network Management Protocol (SNMP)

RFC 1212 Concise MIB definitions

RFC 1213 MIB for network management of TCP/IP-based Internets: MIB-II

RFC 1215 Convention for defining traps for use with the SNMP

RFC 1227 SNMP MUX protocol and MIB

RFC 1239 Standard MIB

RFC 1724 RIPv2 MIB extension

RFC 2011 SNMPv2 MIB for IP using SMIv2

RFC 2012 SNMPv2 MIB for TCP using SMIv2

RFC 2013 SNMPv2 MIB for UDP using SMIv2

RFC 2096 IP forwarding table MIB

RFC 2578 Structure of Management Information v2 (SMIv2)

RFC 2579 Textual conventions for SMIv2

RFC 2580 Conformance statements for SMIv2

RFC 2674 Definitions of managed objects for bridges with traffic classes, multicast filtering and VLAN extensions

RFC 2741 Agent extensibility (AgentX) protocol

RFC 2787 Definitions of managed objects for VRRP

RFC 2819 RMON MIB (groups 1,2,3 and 9)

RFC 2863 Interfaces group MIB

RFC 3164 Syslog protocol

RFC 3176 sFlow: a method for monitoring traffic in switched and routed networks

RFC 3411 An architecture for describing SNMP management frameworks

RFC 3412 Message processing and dispatching for the SNMP

RFC 3413 SNMP applications

RFC 3414 User-based Security Model (USM) for SNMPv3

RFC 3415 View-based Access Control Model (VACM) for SNMP

RFC 3416 Version 2 of the protocol operations for the SNMP

RFC 3417 Transport mappings for the SNMP

RFC 3418 MIB for SNMP

RFC 3621 Power over Ethernet (PoE) MIB

RFC 3635 Definitions of managed objects for the Ethernet-like interface types

RFC 3636 IEEE 802.3 MAU MIB

RFC 4188 Definitions of managed objects for bridges

- RFC 4318 Definitions of managed objects for bridges with RSTP
- RFC 4560 Definitions of managed objects for remote ping, traceroute and lookup operations
- RFC 7011 IPFIX: a method of exporting IP flow data in a network for analysis

Multicast Support

Bootstrap Router (BSR) mechanism for PIM-SM

IGMP query solicitation

IGMP snooping (IGMPv1, v2 and v3)

IGMP snooping fast-leave

IGMP/MLD multicast forwarding (IGMP/MLD proxy)

MLD snooping (MLDv1 and v2)

RFC 1112 Host extensions for IP multicasting (IGMPv1)

RFC 2236 Internet Group Management Protocol v2 (IGMPv2)

RFC 2710 Multicast Listener Discovery (MLD) for IPv6

RFC 2715 Interoperability rules for multicast routing protocols

RFC 3306 Unicast-prefix-based IPv6 multicast addresses

RFC 3376 IGMPv3

RFC 3973 PIM Dense Mode (DM)

RFC 4541 IGMP and MLD snooping switches

RFC 4601 Protocol Independent Multicast - Sparse Mode (PIM-SM): protocol specification (revised)

Open Shortest Path First (OSPF)

OSPF link-local signaling

OSPF MD5 authentication

OSPF restart signaling

Out-of-band LSDB resync

RFC 1245 OSPF protocol analysis

RFC 1246 Experience with the OSPF protocol

RFC 1370 Applicability statement for OSPF

RFC 1765 OSPF database overflow

RFC 2328 OSPFv2

RFC 2370 OSPF opaque LSA option

RFC 2740 OSPFv3 for IPv6

RFC 3101 OSPF Not-So-Stubby Area (NSSA) option

RFC 3509 Alternative implementations of OSPF area border routers

RFC 3623 Graceful OSPF restart

RFC 3630 Traffic engineering extensions to OSPF

Quality of Service (QoS)

IEEE 802.1p Priority tagging

IEEE 802.1Qaz Enhanced Transmission Selection (ETS)

RFC 2211 Specification of the controlled-load network element service

RFC 2474 DiffServ precedence for eight queues/port

RFC 2475 DiffServ architecture

RFC 2597 DiffServ Assured Forwarding (AF)

RFC 2697 A single-rate three-color marker

RFC 2698 A two-rate three-color marker

RFC 3246 DiffServ Expedited Forwarding (EF)

Resiliency Features

IEEE 802.1AX Link aggregation (static and LACP)

IEEE 802.1D MAC bridges

IEEE 802.1s Multiple Spanning Tree Protocol (MSTP)

IEEE 802.1w Rapid Spanning Tree Protocol (RSTP)

IEEE 802.3ad Static and dynamic link aggregation

Routing Information Protocol (RIP)

RFC 1058 Routing Information Protocol (RIP)

RFC 2081 RIPv2 protocol applicability statement

RFC 2453 RIPv2

Security Features

SSH remote login

SSLv2 and SSLv3

TACACS+ accounting, authentication and authorisation (AAA)

IEEE 802.1X Authentication protocols (TLS, TTLS, PEAP and MD5)

IEEE 802.1X Multi-suplicant authentication

IEEE 802.1X Port-based network access control

RFC 2818 HTTP over TLS ("HTTPS")

RFC 3546 Transport Layer Security (TLS) extensions

RFC 4251 Secure Shell (SSHv2) protocol architecture

RFC 4252	Secure Shell (SSHv2) authentication protocol	RFC 1091	Telnet terminal-type option	VLAN Support Generic VLAN Registration Protocol (GVRP) IEEE 802.1Q Virtual LAN (VLAN) bridges IEEE 802.1v VLAN classification by protocol and port IEEE 802.3acVLAN tagging
RFC 4253	Secure Shell (SSHv2) transport layer protocol	RFC 1350	Trivial File Transfer Protocol (TFTP)	
RFC 4254	Secure Shell (SSHv2) connection protocol	RFC 1985	SMTP service extension	
RFC 8446	Transport Layer Security (TLS) v1.3	RFC 2049	MIME	
RFC 8894	Simple Certificate Enrollment Protocol (SCEP)	RFC 2132	DHCP options and BootP vendor extensions	
Services		RFC 2616	Hypertext Transfer Protocol - HTTP/1.1	Voice over IP (VoIP) LLDP-MED ANSI/TIA-1057 Voice VLAN
RFC 854	Telnet protocol specification	RFC 2821	Simple Mail Transfer Protocol (SMTP)	
RFC 855	Telnet option specifications	RFC 2822	Internet message format	
RFC 857	Telnet echo option	RFC 4330	Simple Network Time Protocol (SNTP) version 4	
RFC 858	Telnet suppress go ahead option	RFC 5905	Network Time Protocol (NTP) version 4	

Feature Licenses

	Description	Includes
AT-FL-G98EM-01	GS980EM Premium license	■ OSPF (128 routes) ■ PIMv4-SM, DM, and SSM ■ RIPng (256 routes) ■ STATIC (128 Routes) ■ EPSR Master

ORDERING INFORMATION

AT-GS980EM/10H	8-port 10/100/1000T PoE++ switch with 2 SFP ports, and 3 external PSU ports ²
AT-GS980EM/11PT	8-port 10/100/1000T PoE+ switch with 2 SFP ports, one AC adapter port ³ , and one PoE-in port ⁴ (supporting PD and PoE pass-through)
AT-PWR300-xx	300W PoE power supply (for GS980EM/10H and x320-10GH switches)

Where xx =

- 10 forUS power cord
- 20 for no power cord
- 30 for UK power cord
- 40 for AU power cord
- 50 for EU power cord

² PWR300 power supplies for the GS980EM/10H must be ordered separately
³ The xGS980EM/11GPT ships with an AC power adapter
⁴ The xGS980EM/11GP can be powered by PoE from 30W (class 4) to 90W (class 8)

Accessories

1000Mbps SFP Modules			
AT-SPTX/I	100 m, 10/100/1000T SFP, RJ-45	AT-SPBD20-14	1000BX GbE Bi-Di (1490 nm Tx, 1310 nm Rx) fiber up to 20 km
AT-SPSX/I	1000SX GbE multi-mode 850 nm fiber up to 550 m	AT-SPBD40-13/I	1000LX GbE single-mode Bi-Di (1310 nm Tx, 1490 nm Rx) fiber up to 40 km, industrial temperature
AT-SPLX10/I	1000LX GbE single-mode 1310 nm fiber up to 10 km industrial temperature	AT-SPBD40-14/I	1000LX GbE single-mode Bi-Di (1490 nm Tx, 1310 nm Rx) fiber up to 40 km, industrial temperature
AT-SPBD20-13	1000BX GbE Bi-Di (1310 nm Tx, 1490 nm Rx) fiber up to 20 km		